

Jim Pham
Technology Infrastructure and Student Access
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Chapter four describes the importance of campus technology infrastructure and student access. Moreover, it is about the impact of purchasing, implementing, maintaining, integrating, and accessing the technology within the institutional fabric. Today, technology helps businesses operate with greater efficiency and productivity, and, like many businesses, colleges and universities have used the same technology to improve communication and collaboration among faculty members, students, and other college administrators, admissions departments, and so on. In addition, technology is also enabling a new way to work, learn, and teach. It is already considered an integral part of the college's strategic and operational planning. Such technology infrastructure includes both the physical and human support infrastructure. Consequently, these infrastructures directly impact funding, academic planning, and employee and student access. This chapter raises many important questions about the role of technology and whether it can enhance student learning. More importantly, it is how to fund and support it.

Technology cannot exist without physical infrastructure. It is an element under the technology infrastructure umbrella. Physical infrastructure includes servers, computers, hardware devices, and operating systems. Depending on the type of computer, it is equipped with the appropriate hardware and operating software. A computer machine can be either a client or a server machine and is networked to one or more servers on the network. All computers connected in this way are considered part of the same network. In addition, there are physical networks, routers, bridges, fiber, and Ethernet. Computer machines operate through the operating software that enables them to work (e.g., Windows XP, Microsoft Word, Internet Browser, and so on). For any computer machine or server to work, operating software needs to

be installed. In addition, the infrastructure can include telecommunications links to the outside world and between campuses, including telephone services, videoconferencing equipment and networks, and the Internet.

No physical infrastructure can function without human support. It is considered even more important than the physical infrastructure. It is the people who not only keep things running and up to date but also provide help and instructions to users who must learn to operate the hardware and software. This chapter discusses four levels of human support. The first support level consists of technology infrastructure support staff, who ensure the network and equipment function properly, including installation, operation, and maintenance. The second support level is the educational technology support staff, who are the media production and services staff—interface designers, graphic designers, and so on. The third support level is the instructional design staff, the education services people—instructional design, faculty development, project management, and evaluation. The fourth support level is the subject experts—professors, instructors, and teachers. Both the technology support staff and the educational technology support staff are responsible for the operational aspects. In contrast to the instructional design staff and the subject experts, they are the people responsible for the creation and delivery of technology-based teaching. Teaching and learning through technology are only possible if the human support staff makes it available and enables information to be delivered.

In today's information age, technology plays an important role in delivering, bringing, and connecting people together from anywhere in the world. The Internet and related mobile devices, such as cell phones and PDAs, are becoming essential means for people to communicate. These technologies are also being used extensively in industrial companies.

Many colleges and universities have already implemented and integrated the same kind of technology infrastructure into their institutional fabric. One example of such a technology is the Internet portal solution for students, faculty, and employees (e.g., AIU's Blackboard). Most people use the offered services through technology, unaware of the physical infrastructure and the human support infrastructure, but the reality is that the technology infrastructure makes it possible. Technology infrastructure is a combination of physical and human support infrastructure.

The cost of technology infrastructure is very expensive. In addition, it is not a one-time cost, and it will cost even more in the future. Although technology is an important tool for communication, teaching, and learning, it is also important to recognize that the development and maintenance of the infrastructure are very complex and expensive. Perhaps before implementing such a costly technology infrastructure, institutions need to ask how it will be funded. In addition to rapid technological advances, there is ongoing consideration of replacing or updating the network, equipment, and operating software. Furthermore, there is an additional cost for the technology support staff. This chapter mentions several ways to fund the technology infrastructure. For example, implementation costs are usually funded from the capital budget, and recurrent costs for technology support staff are treated as part of the annual base operations budget.

There is also a question of who can access technology. Academic priorities and strategies for technology use are more important than deciding how to support students' access to technology. There is no question that the administration, staff, faculty, and academic support staff can access technology, but student access is considered an even more important objective for the technology infrastructure. As the chapter put it, an institution can build the most

sophisticated technological infrastructure to support teaching, but if students cannot access it, all the investment is wasted.

In conclusion, technology serves two purposes: 1) communication and collaboration needs and 2) academic needs. It is essential to have operational and strategy planning for the technology infrastructure. The plan should be driven by the college's overall vision and strategy for teaching. In addition, implementation priorities must be set for both the level and the areas of investment. Keep in mind that technology is not a one-time investment, and there is a continuous need to replace and update it. Furthermore, the costs of physical infrastructure and human support are significant expenditure items that need to be reviewed annually and should be part of the budget planning. For this reason, it is necessary for the institution to develop and maintain a robust information technology infrastructure and a strategy plan for moving forward.

Reference

Bates, A.W. (2000). *Managing technological change: strategies for college and university leaders*. San Francisco, CA: Jossey-Bass.